Course: ENCN423/627 – Sustainable Energy Technologies

Module: Solar

Practice problems: Solar basics, array sizing calculations

Question 1.

Explain how the photoelectric effect creates mobile charge carriers within an n-p junction of two doped semiconductors

Question 2.

What is the advantage of stacking multiple material layers (such as Perovskite and Silicon) within a single cell?

Question 3.

If a cell's open-circuit voltage was tested in standard conditions to be 55 V, estimate the open-circuit voltage at a cell temperature of -5°C if $k_v = -0.15$ V/°C

Question 4.

Why are inverters necessary for solar PV generation, and how do they achieve this function?

Question 5.

List at least 5 potential loads the support structure of a freestanding PV module needs to be sized for during design

Question 6.

What purpose does a bypass diode serve within a PV array? In what situation(s) would one be useful?

Question 7.

How can agrivoltaics systems increase average crop yields?

Question 8.

If a house which receives 3.8 peak solar hours per day has 10 rooftop PV panels, each with a nominal power of 400 W and a performance ratio of 0.91, how much energy could they generate in a year?

Question 9.

A client is considering adding a floating PV array to their existing hydropower asset, present a brief discussion of the potential pros and cons of this endeavour